

PitchBook Data, Inc.

Nizar Tarhuni Executive Vice President of
Research and Market Intelligence

Paul Condra Global Head of Private
Markets Research

James Ulan Director of Emerging
Technology Research

Institutional Research Group

Analysis



John MacDonagh
Senior Research Analyst,
Carbon & Emissions Tech and
Clean Energy Tech
john.macdonagh@pitchbook.com

pbinstitutionalresearch@pitchbook.com

Publishing

Designed by **Adriana Hansen**

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EMERGING TECH RESEARCH

Tariffs and Supply Chain Challenges in Clean Energy

Tariffs are compounding the existing challenges faced by clean energy supply chains

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Key takeaways

- Clean energy development and adoption are exposed to significant risk due to geographically concentrated supply chains. The falling cost of low-carbon energy technologies such as solar, wind, and lithium batteries has accelerated their adoption. However, their dependence on critical minerals and components from geographically concentrated supply chains—particularly in China and Southeast Asia—creates exposure to geopolitical and operational disruptions.
- US tariffs are impacting the country's ability to develop and adopt clean energy technologies. Recent tariffs from the US are affecting clean energy supply chains, increasing costs, and limiting access to key components for the energy sector. The uncertainty for investors and project developers further complicates supply chain management and long-term planning across a broad range of technologies, including renewables, energy storage, and clean fuel generation.
- VC investment in clean energy will be affected by this uncertainty, but there are also opportunities for VC-backed companies to develop technologies in energy storage, recycling, and mineral extraction. VC investment is fueling innovation and development in areas such as lithium battery recycling, direct lithium extraction, subsurface resource detection, and nonlithium battery chemistries. These emerging technologies have the potential to reduce supply chain dependencies, lower environmental impact, and support the global clean energy transition.
- The geographic concentration of clean energy hardware production, mineral extraction, and mineral processing represents a risk for the global clean energy transition. Production of clean energy hardware—particularly solar modules, batteries, and electrolyzers—is highly concentrated in Asia, and scaling up mineral production and manufacturing from low levels in other regions is costly. Some critical minerals have very few alternative supply chains due to limited viable mineral deposits, and others depend on high-volume processing of other minerals because they are produced as byproducts of other mineral extraction/refining processes.

Overview

The clean energy transition continues to accelerate, driven by falling costs for low-carbon energy technologies; the continued electrification of transportation, manufacturing, and the built environment; higher consumption of energy produced by low-carbon technologies; and increased energy demand from growing datacenter energy requirements. Low-carbon energy technologies are varied, and include:

- Intermittent renewable energy sources such as solar and wind, which are highly mature, with good global availability, a low levelized cost of energy (LCOE), and rapidly developing talent pools. The falling LCOE of these technologies means that in many cases they are the most viable option when adding new energy generation capacity, regardless of low-carbon goals.
- Dispatchable low-carbon energy sources, including renewable sources such as geothermal, hydropower, wave and tidal energy, plus nonrenewable low-carbon power sources such as nuclear fission and fusion.

Renewable energy is increasingly being used as a tool to improve energy independence for countries without their own conventional energy deposits, as renewable assets are not dependent on importing energy commodities. Energy storage is also growing in adoption, allowing easier integration of intermittent renewable energy sources into power networks—smoothing out some of their output variability—and also optimizing energy usage in buildings and facilities for commercial, industrial, and residential applications. Although we see VC-backed companies developing a broad range of technologies for energy storage, new energy storage capacity additions tend to involve lithium battery projects because of the maturity and relatively low cost of the technology.

Even incorporating the need for energy storage to balance the output of intermittent renewables, lower-carbon forms of energy generation can now be cost competitive with fossil fuel facilities. However, the commodity inputs—construction materials, base metals, and critical mineral resources—are different, exposing clean energy technologies to very different supply chain risks than their high-carbon counterparts. The speed of the clean energy transition has caused a rapid increase in demand for energy transition resources, and supply chains for these are often limited by geographically concentrated mineral deposits in addition to highly concentrated processing of these resources. Regionally concentrated mineral extraction and processing can result in supply chain vulnerabilities and bottlenecks with limited alternative sources if problems arise.

Global supply chain challenges may limit the speed at which countries can produce clean energy hardware, but further issues lie in the fact that the production of certain clean energy hardware is also highly geographically concentrated. In particular, the production of solar modules and lithium-ion batteries is highly concentrated in East Asia and Southeast Asia. Wider climate tech projects may also be at risk if the pace of clean energy adoption slows, as many of these technologies are dependent on clean energy to power them. Because these projects involve less

mature technologies, many are less able to pay the high premiums required for clean energy when compared with mature businesses in established sectors such as manufacturing and datacenter operation.

US tariffs

Recently, US President Donald Trump introduced a broad suite of global tariffs across a range of geographies and industries, seeking to boost domestic manufacturing and address perceived trade imbalances. Due to the breadth of the industries and regions covered, these tariffs have a strong potential impact on clean energy supply chains. These impacts are compounded by the strong uncertainty around how the tariffs may be changed. Recently, the US Court of International Trade challenged the sweeping tariffs that Trump introduced on “Liberation Day,” although the Trump administration has announced it will appeal this ruling.¹

Since the Trump administration announced the initial set of tariffs, the details have changed considerably, with tariffs increasing and decreasing in magnitude. Exemptions have been put in place for certain sectors and regions while others have been particularly targeted. Although some tariffs have ultimately been enacted and will almost certainly continue to exist in some form, some of the initial waves of tariff announcements may have been used as a tool to encourage specific trade deals rather than policies that will remain at the stated magnitudes.

While tariffs affecting clean energy supply chains are not a new concept, the breadth of regions targeted makes alternative supply chains limited, and the magnitude of tariffs means that some sectors are not able to handle tariffs by reducing costs in other areas or passing on additional costs to customers. Ultimately, potential supply chain challenges—plus uncertainty around the future state of tariffs—are making VC investors more cautious to invest in certain clean energy technologies. Because of the ever-changing state of US tariff implementation, the effects on specific trade partners will have different impacts on global supply chains, further complicating the task of calculating the cost of tariffs and of finding the most suitable alternative suppliers of minerals and hardware. The uncertainty around tariffs is as much a factor as the tariffs themselves in many cases.

Industries and regions affected

Shared clean energy infrastructure

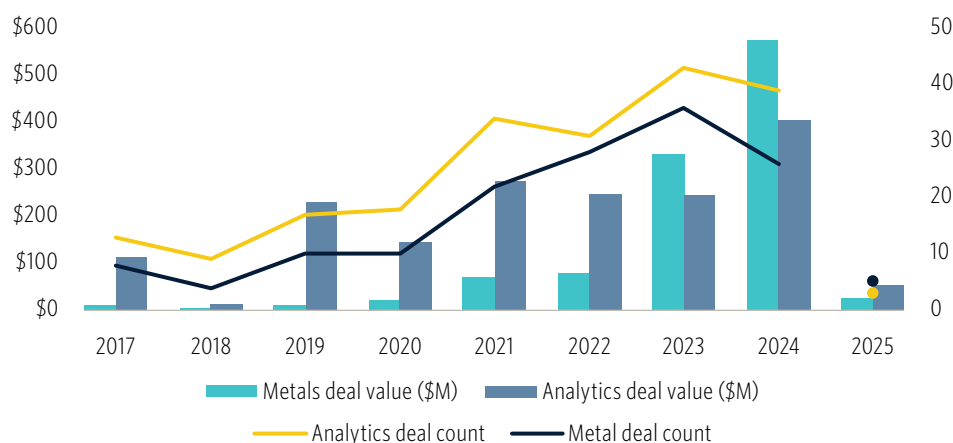
Individual clean energy industries are dependent on specific critical mineral needs, exposing them to unique supply chain risks and challenges. Most energy projects—whether clean or conventional—also depend on a common set of material and component inputs, including basic materials for infrastructure projects, such as cement and steel, and components specific to certain power grids, such as copper and electrical hardware. While these materials and hardware components have long-established markets and supply chains, the US market in particular could see impacts due to recent tariffs, since the country imports significant percentages of cement, steel, and transformers from China, Canada, Mexico, and Europe.

¹: “Trump Tariffs Get to Stay in Place for Now. What Happens Next?” BBC News, Peter Hoskins and Yang Tian, May 30, 2025.

The US relies on Canada and Mexico as the two largest import sources for steel, while China and Germany are also in the top 10.² Mexico and Canada together account for 27% of US cement imports;³ they also account for a significant share of imported electrical transformers.⁴

Copper and steel supply chains can be bolstered to some extent through technologies such as recycling. Metal recycling is a fairly mature space, although \$573.9 million in VC investment went to metal recycling technologies in 2024, marking a high point for the space. Further, VC investment in analytics for recycling and waste sortation can optimize recycling processes, improving the quality and quantity of materials that can be extracted from waste. From 2020 to 2024, there was an average of \$263.5 million in annual VC deal value invested in recycling analytics technologies, mostly covering computer vision, machine learning, and robotics technologies for sorting waste more accurately.

Recycling analytics and metal recycling tech VC deal activity



Source: PitchBook • Geography: Global • As of March 31, 2025

Solar photovoltaics

Solar photovoltaic (PV) panels are a critical component of many regions' energy plans, and the technology and manufacturing capabilities of the space are highly mature. When adding new energy generation capacity, solar is now one of the lowest-cost options,⁵ although manufacturing for the space is geographically concentrated in East Asia and Southeast Asia. Much of this manufacturing occurs outside the VC ecosystem, but we still see large deals from China-based manufacturers of solar hardware, including large deals in the last few years from Huakong Power, Huasun, and Astroenergy. Some of this regional concentration has been driven by China's early investment in the manufacturing and production of solar components, but some is also driven by critical mineral extraction.

2: "United States Steel Imports Report," International Trade Administration, n.d., accessed June 3, 2025

3: "U.S. Cement Industry Statement on Trump Administration's Proposed Tariffs," Portland Cement Association, February 4, 2025.

4: "This Is How Much the U.S. Imports From China, Canada and Mexico," The New York Times, Karl Russell, Lazaro Gamio and Ana Swanson, March 4, 2025.

5: "Lazard LCOE+," Lazard, June 2024.

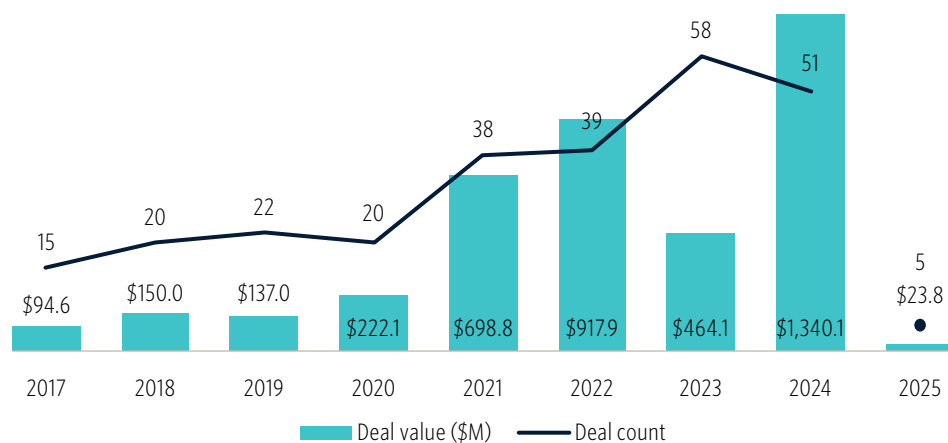
Although different solar PV panel types do not all rely on the same material inputs, some minerals are used in several of the most prevalent designs, including gallium, indium, and tellurium. Gallium extraction is highly concentrated in China, where it occurs as a supporting industry to aluminum mining. This dependency on aluminum mining makes scaling gallium extraction up or down more challenging, and it is also a headwind for other regions looking to develop their own gallium supply chains, as gallium extraction is most economical in regions with large aluminum extraction industries.

For regions outside China, this also makes solar PV hardware production highly dependent on importing specific minerals from China. This is even more challenging because China has placed export controls on certain critical minerals, including gallium—the export of which dropped 75.3% from March 2024 to March 2025.⁶

Tariffs are also a significant influence on the solar PV space. US tariffs for solar hardware are not a new phenomenon, and early tariffs on China-produced panels resulted in higher production in nearby countries Cambodia, Thailand, Vietnam, and Malaysia. Subsequently, US trade officials launched an investigation into solar hardware from these countries that concluded in 2024, although some additional investigations concluded in 2025. The investigations prompted the US to raise tariffs specifically on the solar industries of these four Southeast Asian countries. These countries face tariffs ranging from a 42% tariff on products from Chinese manufacturer JinkoSolar’s Malaysian operations to a 3,521% tariff on hardware from Cambodia, which was a penalty for not cooperating with the investigations.⁷ In the short term, these tariffs have increased the US consumption of solar panels from manufacturers—owned mostly by Chinese companies—based in Laos and Indonesia.⁸

Batteries

Nonlithium battery tech VC deal activity



Source: PitchBook • Geography: Global • As of March 31, 2025

6: "China's Export Controls Are Curbing Critical Mineral Shipments to the World," Reuters, Lewis Jackson and Amy Lv, April 20, 2025.

7: "US Finalizes Tariffs on Southeast Asian Solar Imports," Reuters, April 21, 2025.

8: "US Tariffs, Europe Slowdown Reshape Global Solar Panels Trade," Reuters, Sudarshan Varadhan, May 8, 2025.

When adding new energy storage capacity, lithium-ion batteries remain one of the most widely deployed technologies due to their high energy density, falling costs, and mature manufacturing processes. While other technologies—such as flow batteries, sodium-ion batteries, and nonbattery energy storage—are seeing increased VC investor interest, lithium batteries currently dominate at the grid scale and among commercial, industrial, and residential deployments. This is partly due to the sector sharing a development ecosystem with the electric vehicle (EV) industry, which has resulted in rapid scaling and technological advancement. However, the performance requirements for stationary storage are very different from those of EVs. While EVs prioritize energy density and weight, stationary systems typically focus on cost, safety, and lifespan.

Lithium-ion batteries represent a broad category of battery technology, covering a range of chemistries with different performance characteristics and mineral inputs. Battery chemistries such as lithium nickel manganese cobalt oxides (NMC) are common in EVs due to their high energy density, but they are costly due to their reliance on the critical minerals nickel, cobalt, and manganese. Economically viable deposits of these materials are often concentrated in relatively small regions—cobalt, for instance, is heavily concentrated in Congo, although only about 5% of the mineral is mined by domestically owned mining companies.⁹ Much of the extraction is carried out by Chinese and European mining companies.

Graphite, used extensively for lithium-ion battery anodes, is another key input, and China dominates both natural and synthetic graphite processing. Some emerging technologies are exploring carbon-utilization pathways to produce synthetic graphite, potentially reducing dependence on mined sources. VC-backed companies in the space include CyanoCapture and CO2/Split, although this is a relatively small category within the carbon-utilization space, and most VC-backed companies tend to focus instead on the production of fuels, industrial chemicals, and construction materials.

For stationary energy storage, lithium iron phosphate (LFP) has now become the dominant battery chemistry. LFP batteries offer several advantages: They are less prone to thermal runaway, can have longer working lifespans, and do not require cobalt or nickel inputs, lowering their exposure to supply chain disruption risk.

Despite growing interest in domestic battery production in Europe and the Americas, Asia—particularly China—accounts for the vast majority of lithium battery manufacturing, which makes the sector highly sensitive to trade policy and supply chain disruptions. Attempts to build lithium battery production centers outside of Asia have met challenges. Northvolt, previously described as Europe's answer to its reliance on nondomestic battery production,¹⁰ filed for Chapter 11 protection in the US in 2024 and later filed for bankruptcy in Sweden in 2025, highlighting the challenges faced in trying to rapidly match the years of development enjoyed by China's battery industry. US tariffs on Chinese products, in addition to China's export controls on critical minerals, introduce challenges for clean energy supply chains. Producers of batteries based outside of China face problems

9: "Global Critical Minerals Outlook 2024: Market Review," International Energy Agency, May 2024.

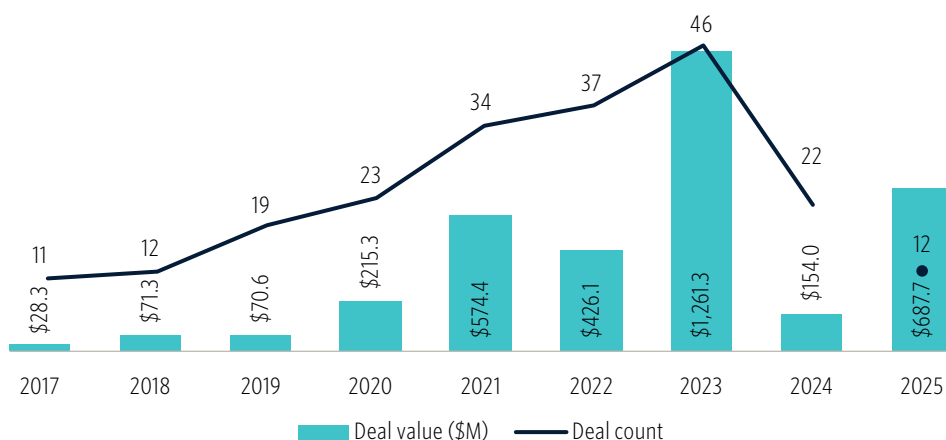
10: "Northvolt Files for Bankruptcy in Sweden," Northvolt, March 12, 2025.

when sourcing raw and processed critical minerals such as lithium, graphite, and manganese, and tariffs hinder clean energy adoption by hiking costs on hardware produced in Asia.

To combat the challenges of battery-material supply chains, VC-backed companies are developing technologies for both recycling end-of-life lithium batteries and for more efficient production of raw materials like lithium. The lithium battery recycling industry has become a critical component of the overall lithium battery space, and in some regions such as the EU, battery producers are required to manage their own recycling programs. Recycling processes focus on recovering a range of valuable materials such as lithium, cobalt, nickel, manganese, and copper. Extracting these minerals from end-of-life batteries can reduce the need for environmentally intensive mining while mitigating some supply chain challenges. VC-backed companies are developing advanced recycling technologies to improve recovery efficiency and reduce operational costs—mostly through lower energy input requirements. The largest VC-backed companies in the space include several North American companies, such as Redwood Materials, Ascend Elements, and Lithion, and the category saw \$3.8 billion in VC deal value in 2023 before falling to \$922.4 million in 2024.

VC-backed companies in the green mining space are developing a range of technologies to better extract lithium from brine and hard rock sources. Startups Lilac Solutions, Summit Nanotech, and ElectraLith are developing direct lithium extraction technologies. Lithium de France and Cornish Lithium are developing geothermal brine extraction technology, which couples heat and/or energy production from geothermal wells with the extraction of minerals from geothermal brine. Green mining processes have the additional benefit of reducing water consumption during the extraction process, which is typically very high.

Green mining VC deal activity



Source: PitchBook • Geography: Global • As of March 31, 2025

The development of nonlithium battery chemistries has increased in recent years, largely driven by growing demand for energy storage and lower costs of production due to use of more abundant mineral resources. This includes flow batteries, metal-air batteries, and sodium-ion batteries, which are less reliant on rare mineral resources, although these technologies generally lack the energy density required for use in EVs and portable devices.

Metal-air batteries in particular, such as zinc-air and iron-air batteries, use widely available, low-cost mineral resources, simplifying supply chain management and reducing overall costs, although round-trip efficiency tends to be lower than most lithium-ion chemistries. Similarly, sodium-ion batteries rely on the far more abundant sodium rather than lithium, making them less vulnerable to supply chain disruptions, although, as with lithium-ion battery production, China continues to lead in sodium-ion battery production, with several major lithium-ion battery manufacturers investing in both chemistries.

Nonlithium battery tech

Sodium-based				
Redox flow				
Zinc/iron				
Other				

Addressing clean energy supply chain challenges

The global clean energy transition faces significant risks due to the geographic concentration of supply chains for mineral resources and key hardware components such as wind turbine parts, solar modules, batteries, and electrolyzers. The production of these components is currently centered in Asia. This concentration represents a vulnerability—disruptions to any of these supply chains could

severely hinder global progress toward decarbonization, and recent geopolitical change has already added to tension between China and the US. While China dominates production of many strategic minerals, recent years have seen strong growth in global supply, helping to keep pace with rising demand. However, price volatility for certain critical minerals remains high, in some cases exceeding that of traditional energy commodities.¹¹

While the most visible disruptions from global supply chain challenges are seen in solar PV and battery technologies, the broader clean energy sector is equally affected by systemic vulnerabilities. Tariffs, the geographic concentration of critical mineral extraction and refining, and heavy reliance on specific regions—particularly Asia—for hardware manufacturing have created bottlenecks in the clean energy industry. For US clean energy development, the geographic concentration of manufacturing, mineral extraction, and mining in regions such as Asia and Europe means that tariffs on these regions leave limited alternative sources for these products. This extends beyond clean energy generation technologies to impact all energy infrastructure projects, as grid modernization and transmission expansion currently depend on access to imported hardware and raw materials.

Scaling up mining and resource extraction outside of established regions such as China and Southeast Asia is challenging. Capital costs for scaling up extraction and refining operations in historically low-volume regions are significantly higher, creating headwinds for regions looking to improve their supply chain resilience.

VC-backed companies are developing more efficient mining, resource extraction, and refining technologies to improve both upstream and downstream resilience.¹² Developments in subsurface resource detection from companies such as KoBold Metals and Ideon Technologies are improving exploration efficiency and reducing environmental impacts, while advances in supply chain management, warehousing, and freight logistics are helping to build more robust and adaptive supply networks.¹³ We also see some companies developing alternative processes for producing specific components of clean energy hardware, such as Niron Magnetics, which develops technology to produce permanent magnets that have key applications in the wind energy industry. Conventional permanent-magnet production supports the energy, automotive, and defense industries and relies on rare earth element extraction. China currently accounts for around 90% of global rare earth magnet production.¹⁴ These technologies—along with those for alternative energy storage and mineral extraction—offer ways to mitigate supply chain risks and continue clean energy adoption in regions that have so far relied on imported hardware and minerals.

11: “Global Critical Minerals Outlook 2025,” International Energy Agency, May 2025.

12: Ibid.

13: For more analysis of recent supply chain dynamics, see our [Q1 2025 Supply Chain Tech VC Trends report](#).

14: “China Hits Back at US Tariffs With Export Controls on Key Rare Earths,” Reuters, Lewis Jackson, et al., April 4, 2025.